

LM09 Analysis of Income Taxes

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1. Introduction

One of the key concepts we will discuss in this learning module is deferred tax assets and liabilities. Deferred tax assets and liabilities are created because of differences between how and when transactions are recognized for financial reporting purposes relative to tax reporting.

2. Differences Between Accounting Profit and Taxable Income

Some common terms related to financial reporting are defined below:

- **Accounting profit:** It is also known as pretax income or earnings before tax (EBT) and appears on the income statement. In simple terms, this is before taxes are calculated. Accounting profit is based on accounting standards.
- **Income tax expense:** Tax expense, or tax benefit, appears on a company's income statement, which is created using financial reporting standards. It is calculated based on the accounting profit (profit before tax) using a given tax rate.
- **Carrying value:** The net value of an asset or liability reported on the balance sheet according to accounting principles.

Some common terms related to tax reporting are defined below:

- **Taxable income:** It is the portion of income that is subject to income taxes under the tax laws where the company is operating.
- **Income tax payable:** Income tax payable is calculated on a company's taxable income using the applicable tax rate. This is the amount that is generally paid to the tax authorities and it appears on the balance sheet. Since it results in a cash outflow, firms minimize taxes payable by showing higher expenses and lower taxable income.
- **Tax Base of an Asset:** Tax base of an asset is the amount that will be deductible for tax purposes in future periods as economic benefits are realized. It is used to calculate tax payable and is analogous to the carrying amount (net book value) concept. Tax base is the amount allocated to asset for tax purposes whereas carrying amount is based on accounting principles.

Why are accounting profit and taxable income different?

Both report income before deducting tax expense, yet they are different because accounting profit is based on accrual method of accounting (revenues reported when earned and expenses when incurred). On the other hand, taxable income is usually based on cash-basis accounting (revenue recognized when cash is collected and expense reported when cash is paid).

Accounting profit and taxable income differ when:

- Revenues and expenses are recognized in one period for accounting purposes and a

different period for tax purposes.

- The carrying amount and tax base of assets/liabilities differ.
- Gain/loss of assets/liabilities in the income statement is different than tax return.
- Some revenues/expenses recognized in the income statement are not considered for tax purposes.

The following table shows the distinction between accounting profit/ taxable income and income tax expense/taxes payable.

Income Statement for Everest Inc.			Tax Return for Everest Inc.		
\$ million	2011	2012	\$ million	2011	2012
Revenue	100	100	Revenue	100	100
Cash Expenses	50	50	Cash Expenses	50	50
Depreciation (SL)	25	25	Depreciation (Acc)	40	10
Accounting profit	25	25	Taxable income	10	40
Income tax expense (40%)	10	10	Taxes payable (40%)	4	16
Profit after tax	15	15	Profit after tax	6	24

Instructor's Note:

The following table summarizes the analogous financial and tax reporting terms:

Financial reporting	Tax Reporting
Accounting profit	Taxable income
Tax expense	Income tax payable
Carrying amount	Tax base

3. Deferred Tax Assets and Liabilities

Deferred tax liabilities

Deferred tax liability (DTL) occurs when income tax expense (financial accounting) is greater than income tax payable. It is a liability because we pay less tax now, thereby creating a liability or an obligation to pay more in the future. Since the tax will be paid later, it is deferred.

Such a situation can happen when:

- Revenue is recognized on income statement before being included on tax return (accrued/unbilled revenue).
- Expenses are tax deductible before being recognized on income statement.

For example, in the sample income statement and tax return shown for Everest Inc, at the end of 2011, the income tax expense (10) is greater than the income tax payable (4), hence a

DTL of 6 (10 - 4) will be recorded on the balance sheet. At the end of 2012, the DTL is reversed and it increases taxes payable by 6.

Deferred Tax Assets

Deferred tax assets (DTA) arise when income tax payable is temporarily greater than income tax expense. In other words, taxable income is higher than accounting profit. Since tax is paid in advance, it is considered an asset; it can be viewed as a prepaid expense.

Such a situation can happen when:

- Revenue is taxed before being recognized on income statement (unearned revenue).
- Expense is recognized on the income statement before being tax deductible.

Consider the following income statement and tax return for Atlas Inc.

Income Statement for Atlas Inc.			Tax Return for Atlas Inc.		
\$ million	2011	2012	\$ million	2011	2012
Revenue	100	100	Revenue	120	80
Cash Expenses	50	50	Cash Expenses	50	50
Accounting profit	50	50	Taxable income	70	30
Income tax expense (40%)	20	20	Taxes payable (40%)	28	12
Profit after tax	30	30	Profit after tax	42	18

At the end of 2011, since the income tax payable (28) is greater than the income tax expense (20), a DTA of 8 (28 - 20) will be recorded on the balance sheet. At the end of 2012, the DTA is reversed and it brings down taxes payable by 8.

Any deferred tax asset or liability is the result of a temporary difference that is expected to reverse in the future. Deferred tax liability reverses when taxes are paid in the future resulting in cash outflows. Similarly, deferred tax asset reverses when tax benefits are realized in the future resulting in lower cash outflows.

Realizability of Deferred Tax Assets

A DTA may be created only if the company expects to be able to realize the economic benefit of the deferred tax asset in the future.

If it is uncertain whether future economic benefits will be realized from a temporary difference (for example, if the company is being liquidated), the temporary difference will not result in the recognition of a DTA.

If a DTA was previously recognized, but now there is sufficient doubt about whether the economic benefits will be realized, then, under IFRS, the DTA would be reversed. Under US

GAAP, a DTA is reduced by creating a valuation allowance (a contra account).

Tax Base of an Asset

Asset tax base is the value of an asset according to tax rules and is used to calculate tax payable. Asset tax base is analogous to carrying amount (net book value).

Example

An asset is purchased for 50 and is depreciated over two years. On the financial statements the depreciation is 25 and 25. According to tax rules the depreciation is 40 and 10. Show the carrying amount and tax base at T=0, T=1, and T=2.

Time period	Carrying Amount (Financial Reporting)	Tax Base (Tax reporting)
T = 0	50	50
T = 1	25	10
T = 2	0	0

Link between Tax Base and DTL

Deferred tax liability = (Carrying amount – Tax base) x Tax rate

Assuming a 40% tax rate for the above example,

At T = 1: DTL = (25 - 10) x 0.4 = 6

At T = 2: DTL = (0 - 0) x 0.4 = 0

Instructor's Note: If the carrying amount and tax base are the same then DTL is 0. If the carrying amount is greater than the tax base, then there will be a deferred tax liability. If carrying amount is less than the tax base, then there will be a deferred tax asset.

Both DTL and DTA should be measured at the tax rate which is expected to apply when the liability is settled (reversed).

Link between Income Tax Expense, Tax Payable, and DTL

Income tax expense = Income tax payable + Change in net DTL

where net DTL = DTL – DTA and change in net DTL is the ending value of net DTL – beginning value of net DTL.

Example

In 2015, the income tax payable for a certain company is 100. During the year, DTL increased from 20 to 25 and DTA increased from 0 to 10. What is the provision for income tax in 2015?

Solution:

ITE = ITP + Δ DTL – Δ DTA = 100 + 5 – 10 = 95

Determining the Tax Base of an Asset

Asset tax base is the amount that will be deductible for tax purposes in future periods as the economic benefits become realized.

Examples:

<i>Item</i>	<i>Carrying Amount</i>	<i>Tax Base</i>	<i>Temporary Difference</i>
An asset is purchased for 50; for year 1 depreciation = 25 on income statement and 40 for tax purposes.	25	10	15
Capitalized development cost = 100 at the start of the year. During the year 30 was amortized. For tax purposes only 25% amortization is allowed.	70	75	-5
Research cost for the year = 100; entire cost was expensed. Tax rules require cost to be spread over 5 years.	0	80	-80
Gross accounts receivable = 100 Provision for doubtful debt = 10%. Tax authorities allow 20%.	90	80	10

Determining the Tax Base of a Liability

The tax base of a liability is the carrying amount of the liability less any amounts that will be deductible for tax purposes in the future.

Example:

<i>Item</i>	<i>Carrying Amount</i>	<i>Tax Base</i>	<i>Temporary Difference</i>
Customer payments received in advance = 50 Amount is taxable.	50	0	50

Since the customer pays 50 in advance. A liability called unearned revenue is created on the accounting side making the carrying amount of the liability to 50. On the tax side, 50 is shown as revenue and taxes are paid for the same. So, tax base is 0 and carrying amount is 50.

Temporary and Permanent Differences between Taxable and Accounting Profit

Permanent differences are differences between tax and financial reporting of revenue (expenses) that will **not** be reversed at some future date. These differences do not give rise to DTLs and DTAs. Examples include:

- Income or expense items not allowed by tax legislation. One example that leads to a permanent difference is when a company incurs a penalty or fine on breaking a civil or criminal law.

- Tax credits for some expenditures that directly reduce taxes.

As no deferred tax item is created for permanent differences, all permanent differences result in a difference between the company's effective tax rate and statutory tax rate.

$$\text{Reported effective tax rate} = \frac{\text{Income tax expense}}{\text{Pretax income}}$$

Example

In 2012, Acme's provision for income tax was 20 against an EBT of 100. In the same year, the tax payable was 25 and the taxable income was 110. What was Acme's effective tax rate for 2012?

Solution:

Effective tax rate = $20/100 = 20\%$

Temporary differences between taxable and accounting profit arise from a difference between the tax base and the carrying amount of assets and liabilities. DTLs and DTAs are only created if there is temporary difference which is expected to reverse in the future.

Some examples of situations that lead to temporary differences in carrying amount and tax base are listed below:

Temporary differences between carrying amount and tax base			
Balance Sheet Item	Carrying amount vs. Tax Base	DTL or DTA	Example
Asset	Carrying amount > Tax Base	DTL	Straight-line depreciation for accounting profit. Accelerated depreciation for taxable profit.
Asset	Carrying amount < Tax Base	DTA	Research cost expensed for accounting profit. Amortized for tax.
Liability	Carrying amount > Tax Base	DTA	Cash from customers before revenue recognition. Cash from customers is taxed.
Liability	Carrying amount < Tax Base	DTL	

Instructor's Note: Remember the first relation for how a DTL is created for an asset. Everything else follows.

4. Corporate Income Tax Rates

There are three types of tax rates that are relevant to analysts:

- Statutory tax rate: The tax rate established by law.

- Effective tax rate: Tax amount reported on the income-tax statement divided by the pre-tax income.
- Cash tax rate: Tax actually paid divided by pre-tax income.

Differences between cash taxes and reported taxes occur due to timing differences between accounting and tax calculations. These differences are reflected as a deferred tax asset or a deferred tax liability on the balance sheet.

Statutory tax rate and effective tax rates can be different for a number of reasons, such as tax credits, adjustments to previous years, withholding tax on dividends, etc.

The effective tax rate of a company that operates in multiple countries is the weighted average of the effective tax rates in each country. If the company is expected to report higher profits in a country with a high tax rate and lower profits in a country with a low tax rate, then its effective tax rate will increase. The effective tax rate is used to forecast earnings on the income statement and the cash tax rate is used to forecast cash flows.

To improve the quality of forecasts, the analyst should adjust for one-time events. Information provided in the notes to the financial statements can be used to identify such events.

If a company's effective tax rates are consistently lower than statutory rates or the effective tax rates reported by competitors, then analysts must scrutinize the reason when forecasting tax expense.

5. Presentation and Disclosure

Measurement of DTL

The treatment of deferred tax liability is discussed below:

- DTL should be classified as debt if the liability is expected to reverse in the future when taxes are paid.
- If it is determined that a DTL will not be reversed, then DTL should be reduced and the amount by which it is reduced should be treated as equity. There is no cash outflow expected in the future. Assume for Period 1 in the example above there is a DTL because of the different depreciation methods used for accounting and tax reporting purposes. Also assume the company is expected to grow at a rate of 30% in the foreseeable future, making the depreciation amounts higher with no reversal in sight. In such cases, the liability will be treated as equity.
- If there is uncertainty about the timing and amount of tax payments, analysts should treat DTLs as neither liabilities nor equity.

Measurement of DTA and Valuation Allowance

If it is determined that the DTA will not be realized because of insufficient future taxable

income to recover the tax asset, then the DTA must be reduced.

Under US GAAP, a DTA is reduced by creating a valuation allowance (a contra account). DTA and net income decrease in the period in which a valuation allowance is established. DTA can be revalued upward by decreasing the valuation allowance which would increase earnings.

Instructor's Note

For the exam, you may think of valuation allowance in terms of depreciation. When depreciation expense goes up, net income comes down. Similarly, if valuation allowance goes up, net income comes down. Depreciation is shown as an expense on the income statement. Similarly, an increase in valuation allowance is shown as a loss on the income statement.

Changes in Income Tax Rates

The measurement of deferred tax assets/liabilities is based on current tax law. But, if there is any subsequent change in tax laws or new income tax rates, then existing deferred tax assets and liabilities must be adjusted to reflect those changes. When income tax rate changes, deferred tax assets and liabilities are calculated based on the new tax rate.

Let us take an example to see what happens to DTL. Assume the carrying amount of an asset is 25 and its tax base is 10. When the tax rate is decreased from 40% to 30%, the effect on DTL is:

$$\text{DTL (old rate)} = (\text{carrying amount} - \text{tax base}) \times \text{tax rate} = (25 - 10) \times 0.4 = 6$$

$$\text{DTL (new rate)} = (25 - 10) \times 0.3 = 4.5$$

DTL changes from 6 to 4.5.

Relationship between tax rate, DTL, and DTA

- Decrease in tax rate reduces both deferred tax liabilities and deferred tax assets.
- Increase in tax rate increases both deferred tax liabilities and deferred tax assets.

Example

Firm A has a net deferred tax liability. The government announces a decrease in the statutory tax rate. Will this change benefit the income statement and balance sheet?

Solution:

If the government announces a decrease in the statutory tax rate, it will cause the net DTL to decrease. A lower tax rate causes the tax expense to decrease and, consequently, the net income and equity to increase.

Example

Rocky Inc. a US-based company, reports the following information:

	2014	2015
Deferred tax asset	100	100
Valuation allowance	25	20
Deferred tax asset, net of valuation allowance	75	80
Deferred tax liability	70	70
Net deferred tax asset	5	10
Tax rate	40%	40%

1. What does the decrease in valuation allowance imply about future profitability?
2. How does the reduction in valuation allowance impact income tax expense and net income?
3. What is the impact on deferred taxes if the tax rate is reduced to 35%?

Solution to 1:

The decrease in valuation allowance implies that the company is more likely to benefit from the deferred tax asset. This is probably because the company expects higher profitability in the future.

Solution to 2:

The reduction in valuation allowance causes the tax expense to be lower and the net income to be higher.

Solution to 3:

If the tax rate is reduced from 40% to 35% that reduces both the deferred tax asset and deferred tax liability. Since the company has a net deferred tax asset, a reduction in the tax rate will cause the net deferred tax asset to be lower. Consequently, the equity value will also decrease.

Instructor's Note: If the valuation allowance is equal to the deferred tax asset, this implies that a company expects no taxable income prior to the expiration of the deferred tax asset.

When a company decreases the valuation allowance, it implies a higher probability that the deferred tax asset will benefit the company.

Summary

LO: Contrast accounting profit, taxable income, taxes payable, and income tax expense and temporary versus permanent differences between accounting profit and taxable income.

Accounting profit: This is the pretax income from the income statement. It is based on accounting standards.

Taxable income: This is income subject to tax. It is based on the tax return.

Taxes payable: It is a liability on the balance sheet calculated using taxable income.

Income tax expense: It is an expense recognized in the income statement, which includes taxes payable and changes in deferred tax assets and liabilities.

$$\text{Income Tax Expense} = \text{Income Tax Payable} + \Delta \text{DTL} - \Delta \text{DTA}$$

Permanent differences => Effective tax rate $\neq$ Statutory Tax rate

Reported effective tax rate = Income tax expense / Pretax Income

Temporary differences between carrying amount and tax base			
Item	Carrying amount vs. Tax Base	DTL/DTA	Example
Asset	Carrying amount > Tax Base	DTL	Straight-line depreciation for accounting profit. Accelerated depreciation for taxable profit.
Asset	Carrying amount < Tax Base	DTA	Research cost expensed for accounting profit. Amortized for tax.
Liability	Carrying amount > Tax Base	DTA	Cash from customers before revenue recognition. Cash from customers is taxed.
Liability	Carrying amount < Tax Base	DTL	

LO: Explain how deferred tax liabilities and assets are created and the factors that determine how a company's deferred tax liabilities and assets should be treated for the purposes of financial analysis.

Deferred tax liabilities are created when:

- Income tax expense is greater than taxes payable.
- This can occur if revenues are recognized on the income statement before being included on the tax return (e.g., credit sales).
- Or expenses are tax deductible before they are recognized on the income statement.

Deferred tax assets are created when:

- Taxes payable are greater than income tax.
- This can occur if revenues are taxable before they are recognized in the income statement (unearned revenue).
- Or when expenses are recognized in the income statement before they are tax deductible.

LO: Calculate, interpret, and contrast an issuer's effective tax rate, statutory tax rate, and cash tax rate.

There are three types of tax rates that are relevant to analysts:

- Statutory tax rate: The tax rate established by law.
- Effective tax rate: Tax amount reported on the income-tax statement divided by the pre-tax income.
- Cash tax rate: Tax actually paid divided by pre-tax income.

Differences between cash taxes and reported taxes occur due to timing differences between accounting and tax calculations. These differences are reflected as a deferred tax asset or a deferred tax liability on the balance sheet.

LO: Analyze disclosures relating to deferred tax items and the effective tax rate reconciliation and explain how information included in these disclosures affects a company's financial statements and financial ratios.

If deferred tax liabilities are expected to reverse in the future they can be classified as liabilities. If they are not expected to reverse in future, they can be classified as equity.

If there is any change in tax laws or new income tax rates, then existing deferred tax assets and liabilities must be adjusted to reflect those changes.

Decrease in tax rate reduces both deferred tax liabilities and deferred tax assets.

Increase in tax rate increases both deferred tax liabilities and deferred tax assets.